



Mathematical
Institute

A Survey on LLM Safety Evaluation and Common Benchmarks

CHENGHAO DONG

Dec. 10th, 2024

Oxford
Mathematics

- ▶ Taxonomy of LLM Safety Topics (Vulnerabilities)
- ▶ Surveys on Large Models' Reasoning and/or Evaluations
 - ▶ Taxonomy of Study Approaches for Reasoning
 - ▶ Taxonomy of Reasoning Tasks
 - ▶ Taxonomy of Evaluations (Benchmarks)
- ▶ Our Taxonomy of Safety Benchmarks
 - ▶ Safety Benchmark Classification Criteria
 - ▶ Statistics
- ▶ Other Related Works
 - ▶ Other Survey
 - ▶ Compression Affecting Reasoning Abilities

Taxonomy of LLM Safety Topics (Vulnerabilities)

A Survey of Safety and Trustworthiness of LLMs through the Lens of V&V (2023)

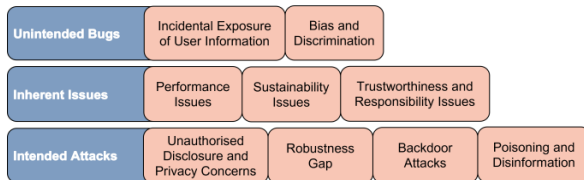


Figure 4: Taxonomy of Vulnerabilities.

- **Unintended Bugs** are made by the developers unconsciously but have serious consequences.

Taxonomy of LLM Safety Topics (Vulnerabilities)

A Survey of Safety and Trustworthiness of LLMs through the Lens of V&V (2023)

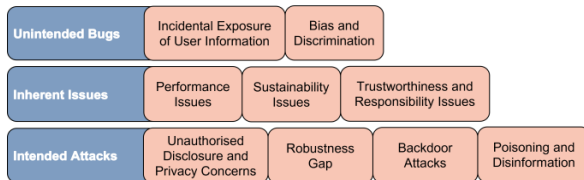


Figure 4: Taxonomy of Vulnerabilities.

- ▶ **Unintended Bugs** are made by the developers unconsciously but have serious consequences.
- ▶ **Inherent Issues** are vulnerabilities that cannot be solved by the LLMs themselves but can be improved with more data and novel training methods.

Taxonomy of LLM Safety Topics (Vulnerabilities)

A Survey of Safety and Trustworthiness of LLMs through the Lens of V&V (2023)

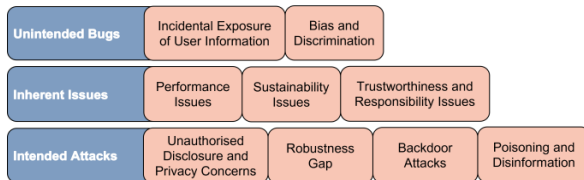


Figure 4: Taxonomy of Vulnerabilities.

- ▶ **Unintended Bugs** are made by the developers unconsciously but have serious consequences.
- ▶ **Inherent Issues** are vulnerabilities that cannot be solved by the LLMs themselves but can be improved with more data and novel training methods.
- ▶ **Intended Attacks** are initiated by malicious attackers.

Taxonomy of LLM Safety Topics (Vulnerabilities)

A Survey of Safety and Trustworthiness of LLMs through the Lens of V&V (2023)



Figure 4: Taxonomy of Vulnerabilities.

- ▶ **Unintended Bugs** are made by the developers unconsciously but have serious consequences.
- ▶ **Inherent Issues** are vulnerabilities that cannot be solved by the LLMs themselves but can be improved with more data and novel training methods.
- ▶ **Intended Attacks** are initiated by malicious attackers.

Surveys on Large Models' Reasoning Abilities

Taxonomy of Study Approaches for Reasoning

Reasoning with Large Language Models, a Survey (2024):
Proposed a taxonomy to classify recent **in-context reasoning** literature based on their **study approaches**. Mainly focuses on studying large models' **mathematical reasoning** abilities.

Stage	Approach	Description	Example Studies
Prompt Generation	Hand-written	Prompts are written by human.	CoT
	External Knowledge	Prompts come from a source that is external to the model, such as another model or a dataset.	Self-ask
	Self-generated	The model itself can be prompted to generate a series of prompts to analyze the problem.	Auto-CoT
Result Evaluation	Self-assessment	The model is evaluated by itself.	Self-verification, Self-consistency
	External Program	The model is evaluated by an external program.	Program-aided-language
	External Model	The model is evaluated by another external model.	Say-can
Reasoning Control	Greedy Selection	The model generate a step once a time and then follows it till the end.	CoT, Auto-CoT, Zero-shot CoT
	Ensemble Strategy	The model generate a set of possible next steps to follow.	Self-consistency, Self-verification
	Control Algorithms	External control algorithms, such as tree-shaped search or reinforcement learning are used to optimize steps.	Tree-of-thoughts, Buffer-of-thoughts

Surveys on Large Models' Reasoning Abilities

Taxonomy of Reasoning Tasks



Mathematical
Institute

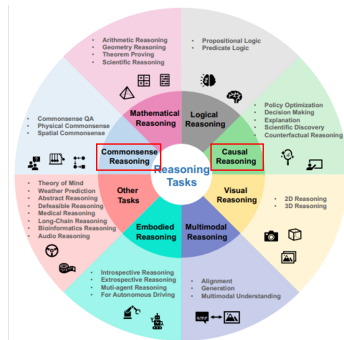
A Survey of Reasoning with Foundation Models (2024): Proposed a taxonomy to classify **reasoning tasks** of LLMs and also provided a series of **benchmarks** used for the evaluation of each task.

Dataset	Choices	Knowledge Types	Questions
Swag Zellers et al. (2018)	4	Temporal, Physical	113,000
PHYRE Bakhtin et al. (2019)	/	Physical	25
HellaSwag Zellers et al. (2019)	4	Temporal, Physical	70,000
WinoGrande Sakaguchi et al. (2021)	2	Social, Physical	44,000
Social IQA Sap et al. (2019)	3	Social	35,350
PIQA Bisk et al. (2020)	2	Physical	21,020
SummEdits Laban et al. (2023)	2	Social	6,348
CCoNS Kondo et al. (2023)	/	Physical	1,112

Table 5: Commonsense Reasoning Benchmark Statistics. Choices: the number of choices for each question; Questions: the number of questions.

Dataset	Tasks	Size
Tübingen cause-effect pairs dataset Mooij et al. (2016)	Causal	108
Neuropathic Pain dataset Tu et al. (2019)	Causal	N/A
Arctic sea ice dataset Huang et al. (2021c)	Causal	N/A
CRASS Froberg and Binder (2021)	Causal	275

Table 9: Causal Reasoning Benchmark Statistics.



Surveys on Large Models' Reasoning Abilities

Taxonomy of Evaluations (Benchmarks)

What to Evaluate	Subclasses
NLP	Natural language understanding
	Reasoning
	Natural language generation
	Multilingual
Robustness / Ethics / Biases / Trustworthiness	Factuality (Hallucination)
	Robustness
	Ethics and biases
	Trustworthiness
Social science	N/A
Natural science & engineering	Mathematics
	General science
	Engineering
Medical applications	Medical queries
	Medical examination
	Medical assistants
Agent applications	N/A
Other applications	Search and recommendation
	Personality testing
	Specific tasks

- *A Survey on Evaluation of Large Language Models (2023):*
Proposed a taxonomy for **benchmarks** of LLMs evaluations.

Surveys on Large Models' Reasoning Abilities

Taxonomy of Evaluations (Benchmarks)

What to Evaluate	Subclasses
NLP	Natural language understanding
	Reasoning
	Natural language generation
	Multilingual
Robustness / Ethics / Biases / Trustworthiness	Factuality (Hallucination)
	Robustness
	Ethics and biases
	Trustworthiness
Social science	N/A
Natural science & engineering	Mathematics
	General science
	Engineering
Medical applications	Medical queries
	Medical examination
	Medical assistants
Agent applications	N/A
Other applications	Search and recommendation
	Personality testing
	Specific tasks

- ▶ *A Survey on Evaluation of Large Language Models (2023):*
Proposed a taxonomy for **benchmarks** of LLMs evaluations.
- ▶ Specific Benchmarks for **Safety Evaluation**:
SafetyBench and **Cvalues** (Chinese)

Our Taxonomy of Safety Benchmarks

Safety Benchmark Classification Criteria



Mathematical
Institute

Classification	Target on	Corresponding Vulnerability	
		Layer 2	Layer 1
Reasoning	Commonsense	Performance	Inherent issues
	Causal		
	Math (Complexity)		
	Medical		
Factuality	N/A		
Comprehension	N/A		
Trustworthiness	Model	Trustworthiness and Responsibility	
Robustness	Adversarial Attack	Robustness Gaps	Intended Attacks
	OOD		
Toxicity	N/A	Bias and Discrimination	Unintended Bugs
Comprehensive	Safety	/	
	Comprehension & Reasoning		
	Reasoning & Safety		

A total of 40 different benchmarks were selected from the survey discussed previously.

Our Taxonomy of Safety Benchmarks

Safety Benchmark Classification Criteria



Mathematical
Institute

Benchmark & Dataset	Classification	Target on	Focus	Choices Given	Size	Time	Source
CQA	Reasoning	Commonsense	N/A	T	12000	2019	8.3.1
NPHardEval	Reasoning	Math	Problem Complexity	F	N/A	2023	N/A
MultiMedQA	Reasoning	Medical	Medical	T	N/A	2022	8.3.2 [117]
CRASS	Reasoning	Causal	N/A	T	275	2021	8.3.1
CELLO	Comprehension	N/A	Complex Instructions	F	523	2023	8.3.2 [66]
TruthfulQA	Factuality	N/A	N/A	F	817	2022	8.3.2 [119]
EmotionBench	Trustworthiness	Model	Emotion Mimicry	T	36	2023	8.3.2 [76]
AdvGLUE++	Robustness	Adversarial Attack	N/A	F	42017	2023	8.3.2 [201]
GLUE-X	Robustness	OOD	N/A	F	6404940	2023	8.3.2 [234]
CHBias (Chinese)	Toxicity	N/A	Bias	T	4800	2023	8.3.2 [256]
SafetyBench	Comprehensive	Safety	N/A	T	11435	2023	8.3.2 [254]
Cvalues (Chinese)	Comprehensive	Safety	N/A	T	4312	2023	8.3.2 [230]
DecodingTrust	Comprehensive	Safety	N/A	F	N/A	2023	8.3.2 [201]
KoLA	Comprehensive	Comprehension & Reasoning	N/A	F	N/A	2023	8.3.2 [236]
BIG-bench	Comprehensive	Reasoning & Safety	N/A	F	N/A	2022	8.3.2 [182]

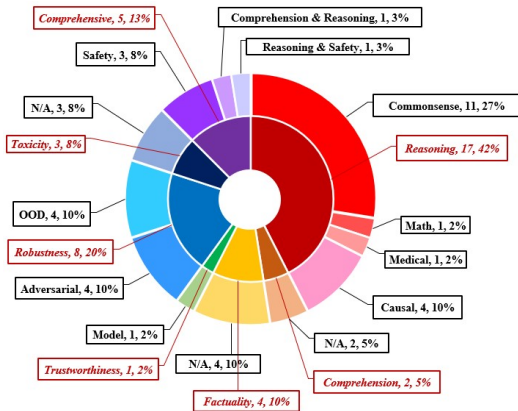
15 Samples of Selected Benchmarks from Each Category.

Our Taxonomy of Safety Benchmarks

Statistics



Mathematical
Institute



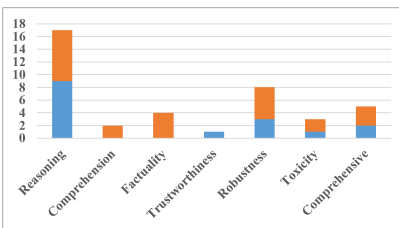
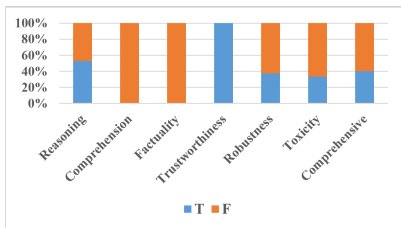
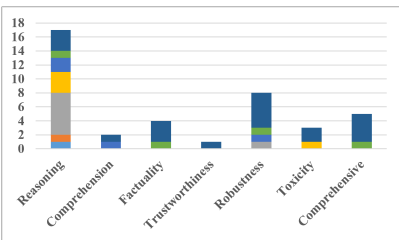
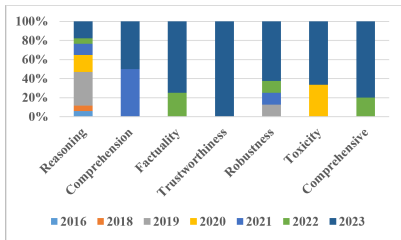
Distribution of 40 selected benchmarks.

Our Taxonomy of Safety Benchmarks

Statistics



Mathematical
Institute



- ▶ More on **Trustworthiness and Robustness**: *DECODINGTRUST: A Comprehensive Assessment of Trustworthiness in GPT Models (2023)*
- ▶ More on **Hallucination and Factuality**: *UHGEval: Benchmarking the Hallucination of Chinese LLMs via Unconstrained Generation (2023)*
- ▶ More on LLM **Comprehension and Semantic Inference**: *KoLA: Carefully Benchmarking World Knowledge of LLMs (2023)*
- ▶ More on LLM **Comprehensive Evaluations**: *Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models (2022)* and *Holistic Evaluation of LMs (2022)*

How Numerical Precision Affects Mathematical Reasoning Capabilities of LLMs (2024): **Quantization** can significantly influence the accuracy of Transformer-based LLMs in basic **mathematical reasoning tasks**. Their results show that

- ▶ Transformers operating with low numerical precision (int8 or int4), fail to address arithmetic tasks like iterated addition and integer multiplication, unless the model size grows **super-polynomially** with respect to the input length.
- ▶ Transformers with standard numerical precision, such as float32, can efficiently handle target arithmetic tasks with significantly smaller model sizes.